An EEG-Based Positive Feedback Mechanism for VR Mindfulness Meditation to Improve Emotion Regulation

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***Abstract*—Virtual reality (VR) mindfulness meditation has emerged as a prominent emotion regulation strategy in recent years. Current research often seeks to enhance meditation effectiveness through biofeedback and overlooks the trajectory of emotional changes and the changing needs during regulation. In this study, we propose an electroencephalography (EEG)–based positive feedback mechanism for VR mindfulness meditation aimed at optimizing the effects of emotion regulation. This mechanism consists of three modules: 1) EEG-based emotional state computation; 2) process-based relaxation assessment; and 3) adaptive positive decision feedback. Collectively, these components form a computation-assessment-feedback closed-loop system that objectively quantifies emotions while enabling real-time decision adjustments based on emotional trends, thereby enhancing user engagement and emotion regulation efficacy through personalized feedback. The contribution of the proposed feedback mechanism was evaluated through a randomized controlled trial (N = 36). The results indicated that both physiological measures and self-reported relaxation significantly increased when compared to interventions without feedback. These findings validate that the EEG-based positive feedback mechanism effectively enhances emotion regulation while providing additional insights into improving both the engagement and effectiveness within digital mental health interventions.**

***Index Terms*—Electroencephalography (EEG), emotion regulation, mindfulness meditation, positive feedback, virtual reality (VR).**

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I. Introduction

MINDFULNESS meditation can be broadly defined as the practice or technique of cultivating mindfulness [1], [2], [3]. Numerous studies have confirmed the positive effects of mindfulness meditation in managing stress [4], [5], [6], relieving anxiety [7], [8], and improving brain function [9]. Virtual reality (VR) technology has great potential to support mindfulness meditation. Its immersive feature can enhance presence, reduce distractions, and boost users' focus on their sensations [10], [11], [12]. Currently, empirical studies on VR mindfulness meditation primarily focus on VR technology and treatment outcomes, with less attention paid to users' feelings and emotional states. This gap leads to decreased user compliance with interventions, thereby limiting the effectiveness of emotion regulation.

In recent years, the use of physiological measurements for the objective and accurate assessment of individual emotional states has been widely applied. Some VR meditation systems have incorporated physiological markers to detect emotional states and form biofeedback [10], [13]. This integration makes meditation more accessible [14], promotes longer-term training adherence [15], and enhances the self-improvement and stress-management capabilities of meditation [16], [17], [18]. However, these studies predominantly focus on the current emotional state and the real-time nature of feedback while neglecting the overall trend in users' emotional changes during meditation. Furthermore, the optimizing effect of a single form of feedback on emotion regulation appears to be limited.

In this study, electroencephalography (EEG) is integrated into VR mindfulness meditation to propose an EEG-based positive feedback mechanism, aiming to enhance the efficacy of emotion regulation. The mechanism objectively quantifies emotional states by capturing users' EEG data and comprehensively assesses the degree of relaxation based on emotion scores over a specified period. It infers the needs at different stages of emotion regulation and adaptively generates various forms

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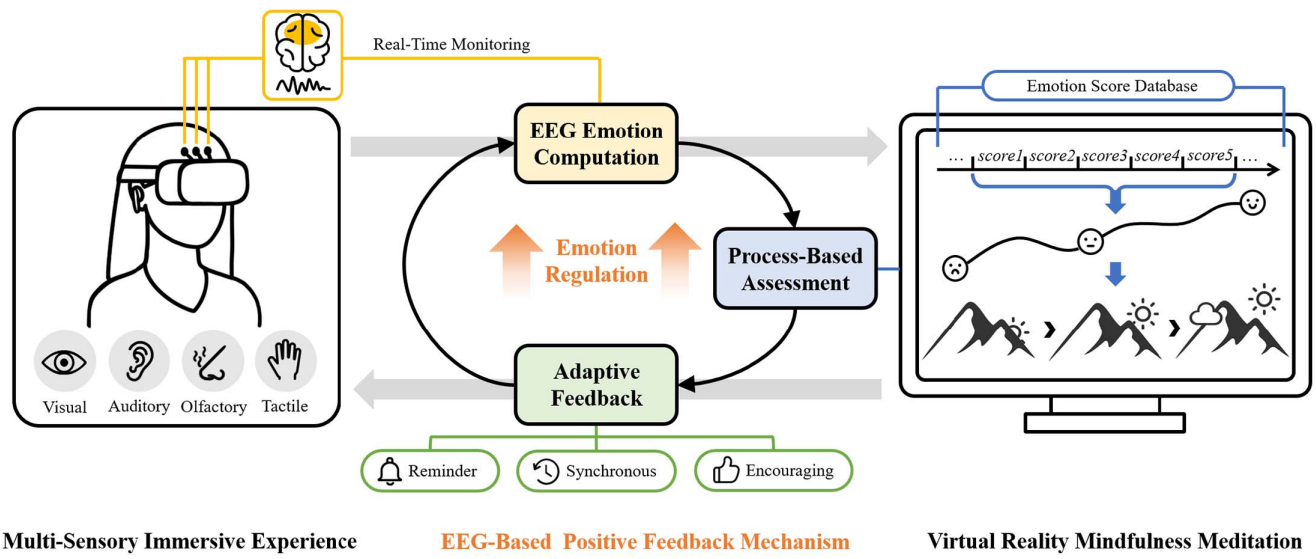

Fig. 1. Overview of the positive interactive closed-loop.

of positive feedback. This process forms a positive feedback closed-loop of "computation–assessment–feedback" (Fig. 1). We design two mindfulness meditation courses and develop corresponding VR scenes while embedding the feedback mechanism to create an interaction prototype. A randomized controlled trial (RCT) is conducted in a laboratory setting involving 36 university students seeking emotional relief to assess the effectiveness of our proposed mechanism. The contributions of this study are as follows:

1) *EEG-based positive feedback mechanism:* Adaptive closed-loop real-time responsiveness to user EEG data facilitates more precise interventions. The principle of positive feedback drives the system to continuously evolve in a positive direction. Consequently, human–computer synergistic interaction improves the emotion regulation capabilities of both parties.
2) *Process-based relaxation assessment:* The degree of relaxation is thoroughly assessed through the continuous monitoring of emotional states, enabling a comprehensive grasp of the trends in emotional changes. This process generates more accurate and universally applicable feedback indicators, thereby providing a reliable foundation for subsequent feedback.
3) *Adaptive positive decision feedback:* Based on the user's emotional state and needs at different stages of emotion regulation, the feedback strategy is flexibly adjusted. Diverse feedback forms are utilized to induce positive emotions. Personalized experiences enhance user engagement and satisfaction, thereby optimizing the effectiveness of intervention.

II. Related Work

A. Quantifying Emotion in VR Using EEG

EEG is a noninvasive measure of the electrical activity generated by neurons in the brain [19], which effectively captures the electrical activity of the central nervous system. Research evidence shows that EEG is closely related to emotional states [20]. It undergoes significant changes in emotions at different valence levels [21], [22], [23], [24]. EEG has been widely used in studies of emotional tasks [25], [26], detection of affective or mental disorders [27], [28], [29], [30], [31], assessment of relaxation [32] and meditation levels [13]. Currently, the field of EEG-based emotion recognition is focused on improving accuracy, and substantial progress has been made. For instance, the optimization of multimodal fusion and data processing algorithms enables emotion recognition to meet cross-cultural and cross-individual needs [33], [34], [35]. Meanwhile, EEG has excellent temporal resolution [36] and multidimensional features, which can complete the simultaneous acquisition of EEG signals and a comprehensive description of emotional processes in VR emotion induction. In addition, the characteristics of EEG, such as noninvasive, economical, and portable, are conducive to controlling the cost and improving the ubiquity when performing emotion monitoring in VR [37]. Based on these advantages, EEG has gradually become an important tool for quantitative research on emotions in combination with VR [38], [39]. However, most empirical studies have focused on the domains of emotion recognition and affective computation [20], [40] and lack applications in emotion regulation.

B. VR Meditation Study Incorporating Biofeedback

Many studies have incorporated bioresponsive components into VR meditation systems to enable biofeedback [10]. Most of these biofeedback presentations are either a visualization of the user's meditative behavior (using the proximity of a cloud to reflect the user's breathing in [41]) or a gamification mechanism (Relaworld [42] uses neurofeedback to control the user's floating in a VR meditation system), and they are real-time feedback of physiological states, which can be used to effectively train the relevant behaviors. If the purpose is emotion regulation, this real-time biofeedback is not applicable in VR

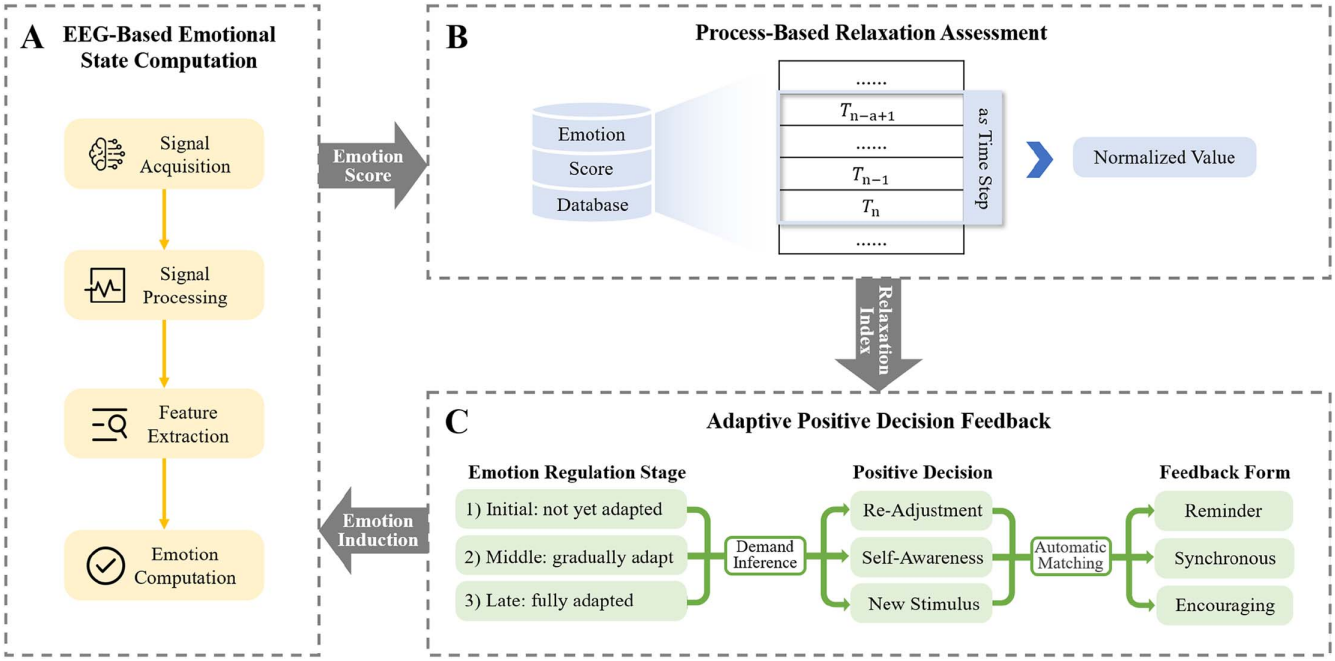

Fig. 2. Framework for EEG-based positive feedback mechanism.

mindfulness meditation, because emotion regulation is characterized by physiological underpinnings, behavioral responses, as well as affective and cognitive correlates [43], [44]. In some other studies, the emotional state at a certain moment does not truly reflect the user's emotional changes, and because physiological signals are susceptible to interference, slight fluctuations can cause a bias in the judgment of the emotional state. In addition, current VR meditation systems that introduce biofeedback provide a single form of feedback, most of which map meditation-related physiological parameters to the VR scene so that they manipulate or influence the visual presentation of the VR elements, e.g., Choo et al. [15] used EEG to track the user's meditation scores and provide real-time visual feedback on the opening of flowers, with the higher the meditation scores, the more flowers blooming and vice versa. However, Braunstein et al. [41] emphasized that emotion regulation goals range from implicit/unconscious to explicit/conscious (e.g., the goal of making oneself happier), and that if only a single unchanging feedback is given during emotion regulation it cannot satisfy all of the user's needs, and cannot achieve the ultimate goal of emotion regulation.

III. Methodology

In this study, EEG biofeedback was integrated with VR mindfulness meditation. The methodology for objectively quantifying emotional states was elaborated upon, and an in-depth exploration of effective feedback design associated with these emotional states was conducted. Therefore, an EEG-based positive feedback mechanism was proposed to enhance the efficacy of emotion regulation within VR mindfulness meditation. This approach offers a referable model for the integration of biofeedback into digital mental health interventions.

The proposed positive feedback mechanism employs real-time monitoring of EEG to objectively quantify emotional states, offering a process-based assessment of an individual's level of relaxation. It adaptively delivers targeted positive decision feedback, thereby enhancing emotion regulation in VR mindfulness meditation. Specifically, as users engage in guided meditation, their physical and mental state changes are continuously recorded by a wearable EEG device that monitors brain activity in real time. This device calculates relevant EEG signals to quantify the degree of relaxation and aligns the user's current needs for emotional induction with their stage of emotion regulation. Consequently, it provides varying forms of positive audiovisual feedback aimed at facilitating mental healing and further promoting psychological relaxation—particularly activating brain regions associated with relaxation. Under this mechanism, a positively facilitated interaction closed-loop is established between the user and the VR system, resulting in an integrative enhancement of emotion regulation capabilities for both parties involved. Fig. 2 illustrates the overall framework of the EEG-based positive feedback mechanism along with its closed-loop formation.

A. EEG-Based Emotional State Computation

The computation of emotional state based on EEG is implemented through a four-step process. First, EEG signals can be acquired from the scalp via EEG sensors [45]. Second, the collected EEG signals undergo processing to enhance the signal-to-noise ratio (SNR). This process includes the application of a median filter algorithm to remove baseline drift, a notch filter to eliminate power-frequency noise, a band-pass filter to remove electromyogram (EMG) noise, and a real-time electrooculogram (EOG) removal algorithm to eliminate ocular

artifacts [46]. Third, the desired EEG features are extracted from the relatively clean EEG signals. EEG reflect different emotional states in different frequency bands. Specifically, alpha waves typically increase during relaxation and meditation, beta waves are associated with high alertness and anxiety, and theta waves appear in drowsiness and idling states in adults [47]. Therefore, in this article, the power of EEG signals in the alpha, beta, and theta bands is extracted, and the emotional state is quantified by the formula $\alpha/(\theta+\alpha+\beta)$. The calculated result is normalized within the range of -1 to 1 to obtain an *emotion score*, where a higher value indicates a better emotional state. Finally, the emotional scores are stored in real time and entered into the next module as available parameters.

The user's emotional state was objectively and accurately assessed through the combination of multiple EEG features, resulting in the generation of a reliable *emotion score*. This module serves as the foundation for more effective implementation of intelligent feedback and represents the primary method for optimizing the impact of emotion regulation.

B. Process-Based Relaxation Assessment

Emotion regulation is a complex psychological process, and the emotion regulation goal and the emotion change process could change in nature before and after regulation [41]. It is impossible to conclude that a person's emotional state is improving or deteriorating based on their emotional response at a given moment. A more comprehensive analysis of emotion states over a period of time and a comprehensive assessment of emotion change trends are necessary to provide more accurate feedback.

In VR mindfulness meditation, it is essential to ascertain whether the user's emotional state is being alleviated or not. This information is crucial for providing targeted feedback that can facilitate emotional regulation. Accordingly, the frequency of positive emotional states during meditation serves as a quantitative measure of the user's relaxation level. Here, an emotional state is deemed positive when the *emotion score* exceeded 0.2. In other words, the objective is to determine the proportion of ratings above 0.2 within a given time interval. A larger proportion means a higher frequency of positive states during that period, implying a greater degree of relaxation. The appropriate time step must then be determined by a combination of the speed of acquisition of emotion scores and the results of multiple pilot experiments. Since module A transmits an *emotion score* at a rate of once every 2 s, the time step was initially set at $2n$ s. Following several rounds of internal measurements to balance the assessment effect and rate, a time step of 20 s was ultimately identified as optimal. This time step contains a total of 10 data points, comprising the current and previous emotion scores. As the meditation progresses, the time window continues to move, generating a quantitative score of relaxation in real time. This value is then normalized and referred to as the *relaxation index*. The *relaxation index* is stored in real time for subsequent analysis of the user's emotion regulation process. Furthermore, it could be applied as a feedback parameter for the VR system.

The conventional approach to biofeedback involves correlating physiological signals with digital feedback, thereby facilitating direct control over changes in biological response elements. In contrast, our proposed positive feedback mechanism emphasizes the specificity of human emotions and incorporates an additional module for transition. A comprehensive assessment of relaxation levels is conducted based on selected data during the emotion regulation process, which allows for a more nuanced understanding of users' emotional fluctuations and enables the provision of more targeted feedback.

C. Adaptive Positive Decision Feedback

In this module, we aim to provide instructive feedback, which can more consciously guide users towards the expected emotional state. The mapping scheme of the feedback strategy is as follows:

First, the stages of emotion regulation are divided. Tang et al. [48] proposed that mindfulness meditation can be generally divided into three distinct practice stages: Early, Middle, and Advanced, each involving different levels of effort. Therefore, we transfer this theory to brief mindfulness meditation practice, and the process of users' emotion regulation can also be divided into three different stages. Taking the obtained relaxation index as the grading standard, the detailed division strategy is as follows.

1) *Relaxation index* < 0.2. The user has not yet adapted to the guided rhythm and needs to make a great deal of effort to adjust their own state, which indicates they are in the initial stage of emotion regulation.
2) *Relaxation index* ≥ 0.2. The user is gradually adapting to the current environment but still needs to make effort to reduce mind wandering, which indicates they are in the middle stage of emotion regulation.
3) *Relaxation index* > 0.8. The user has fully adapted to the meditation practice and can effortlessly exhibit a relaxed and natural state, which indicates they are in the late stage of emotion regulation.

Next, corresponding positive decisions are formulated. During the emotion regulation process, users in different stages have various states and goals, so the additional support they need also varies. By inferring the needs of users at the current stage, an appropriate strategy can be determined. The specific classification is as follows.

1) Users in the initial stage require extended periods and supplementary prompts to facilitate the adjustment of their state and the reacquisition of their distracted attention. Mindfulness breathing is an effective way to guide users in self-regulation during mindfulness meditation [49]. Users can quickly calm down by readjusting their breathing.
2) Users in the middle stage must monitor changes in their emotional state in order to make deliberate adjustments. At this point, it is essential to visualize the user's relaxation level to facilitate the real-time regulation of their emotional state, enhance their attention to their own sensations, and promote greater self-control through enhanced self-awareness.

(a) (b)

Fig. 3. Renderings of two VR scenes. (a) Flower Companion. (b) Yuhua River.

3) Users in the late stage need a more diverse range of stimuli that can elicit positive emotions and enhance the level of relaxation. Thus, it is essential to provide novel stimuli to enhance the user experience and further stimulate the generation of positive emotions.

Finally, these positive decisions are presented to users in different forms of feedback. Based on the nature of various positive decisions, appropriate carriers of expression are selected to integrate guiding opinions into the meditation process and VR scenarios. With the *relaxation index* as the feedback indicator, three forms of feedback are automatically matched as follows.

1) *Reminder feedback*: In traditional feedback, reminder feedback is used to inform subjects that their current performance has not met the preset standard or is subpar, which can enhance the average attention level [50]. This module borrows this form but eliminates its negative connotations. Instead of signaling poor performance to the user, we reintroduce a guidance for mindfulness breathing, which serves as a reminder and provides assistance.
2) *Synchronous feedback*: The *relaxation index* is mapped to the attributes of VR elements, and these biological response components change along with the user's emotions. The subtle changes of VR elements achieves a synchronized visualization of the user's emotional state, allowing for more intuitive self-awareness and fostering motivation and focus [51], [52], [53].
3) *Encouraging feedback*. Encouraging feedback is used to inform subjects that their current performance has met or exceeded the preset standard [50]. Similar to the transformation of reminder feedback, instead of directly showing their performance, we introduce new audio-visual stimuli into the VR scenario. This approach is analogous to the reward mechanism in games, which encourages users to keep relaxing.

This module adaptively generates customized feedback according to the goals and needs of emotion regulation, and the feedback mapping strategy is more interpretable. Importantly, all feedback is designed to be positive, devoid of any negative connotations or punitive elements. Even if the user's current relaxation level is not satisfactory, we will still provide positive guidance. This approach can help users boost their confidence and adhere to the practice better.

IV. Interaction Prototype

To support the feasibility of the proposed feedback mechanism and related modules in this study, we develop an interaction prototype system for VR mindfulness meditation that includes two distinct exercises. Subsequently, the EEG-based positive feedback mechanism is integrated to complete the association between emotional states and VR scenes.

A. Two VR Mindfulness Meditation Exercises

Using the interaction design model for VR mindfulness meditation as a reference [54], we developed two meditation exercises with distinct themes. The first exercise, *Flower Companion*, focuses on flower meditation, which alleviates physical and mental stress through the visualization and sensory experience of flowers. The setting is established on a sunny afternoon, where several giant tulips bloom amidst soft grass [Fig. 3(a)]. Concurrently, floral fragrances are employed to provide olfactory stimulation and enhance the immersive experience. The second exercise, *Yuhua River*, exemplifies rainy day meditation, promoting mental healing through the soothing white noise of rain. We create a peaceful and comfortable atmosphere reminiscent of a rainy day. Users can canoe along the river while observing nature and listening to the sound of rain in pursuit of inner peace [Fig. 3(b)]. Additionally, tactile stimulation is simulated through a small fan that mimics the sensation of breeze.

B. Incorporation of EEG-Based Positive Feedback Mechanism

The EEG monitoring device is externally connected to the VR system in order to construct the feedback mechanism. For these two VR scenarios, we select different virtual elements as the feedback components for synchronous feedback and encouraging feedback, respectively. In *Flower Companion*, flowers are selected as the biological response element in the

synchronous feedback. The user's relaxation level affects the blooming of flowers in real time. The higher the *relaxation index*, the more flowers bloom around. The sound of insects and birds, as well as the sunshine spilling, are selected as new stimuli to enrich the audio-visual experience. In *Yuhua River*, the colorful light spots are selected for synchronous feedback. This records the process of users' emotion regulation through the aggregation and rise of the light spots. Sunshine spilling is chosen as an encouraging feedback element to bring a pleasant experience.

V. Experiment

The objective of this experiment was twofold: first, to evaluate the effectiveness of the EEG-based positive feedback mechanism; and second, to ascertain the feasibility of related modules within this mechanism. A randomized controlled trial (RCT) was conducted to investigate the impact of feedback on emotion regulation.

A. Participants

At this juncture of the study, the following inclusion criteria were implemented, as the objective was to recruit healthy individuals exhibiting mild anxiety symptoms.

1) Adults aged 18 years and older.
2) No physical or mental illnesses and no history of VR-induced vertigo.
3) A score of 6 or higher on the Neuroticism subscale of the Eysenck personality questionnaire-revised short scale (EPQR-S) [55]. Individuals with high neuroticism are characterized by emotional instability, proneness to anxiety, and tension [56], [57], [58].
4) A score ranging from 8 to 25 on the Beck Anxiety Inventory (BAI) [59]. This range identifies individuals experiencing mild to moderate anxiety levels, which have not reached clinical significance [60], [61], [62], [63]. The experiment was conducted in accordance with the ethical standards set forth by the Institutional Review Board and approved by the university's ethics committee. All participants were required to sign an informed consent form prior to the experiment and were duly rewarded at the end.

Eventually, 36 university students (female: 21; age: 23.61 ± 2.77) were invited to participate in the formal experiment. The participants exhibited variability in their familiarity with mindfulness and meditation practices (no knowledge at all: 12, some knowledge: 18, general knowledge: 6). Similarly, their exposure to and utilization of VR technologies varied (no prior experience: 9, limited exposure: 24, frequent use: 3). Despite these differences, all participants expressed interest in exploring digital emotional relief system. Participants (P) were randomly and equally assigned to experimental Group A (with feedback; P01–P18) and control Group B (without feedback; P19–P36). After this, the participants were given the option to choose one of the scenes of *Flower Companion* and *Yuhua River* to experience according to preference. All subjects completed the experiment as required. Therefore, the data from all 36 subjects were included in the analyses.

Fig. 4. Participant wore an integrated VR-EEG device and Bluetooth headset for VR mindfulness meditation. The experimental environment was set up with the floral fragrance and the fan to provide sensory stimulation.

B. Equipments

An integrated device combining VR with an EEG collector (see Fig. 4) was selected as the experimental equipment. This wearable device, developed independently by the UAIS Lab of Lanzhou University, integrates a VR head-mounted display with pervasive three-electrode EEG collector [64]. This EEG sensor collects EEG data from the prefrontal Fp1, Fpz, and Fp2, and uses the earlobe as the reference electrode, which features low power consumption, low cost, and ease of operation. [64], [65]. The integrated device not only enhances comfort and flexibility but also minimizes data interference caused by friction between devices.

C. Measures

1) Self-Report: To understand the emotional changes at different stages, participants were required to complete questionnaires for self-reporting their emotional states. The visual analog scale (VAS) of "anxious," "nervous," and "worried" were scored on a likert scale from 0 (not at all) to 100 (extremely) [66], [67], in addition to the 6-item state-trait anxiety inventory (STAI) [68] and the short Warwick Edinburgh mental well-being scale (SWEMWBS) [69].

2) Physiology: EEG was monitored in real time using a three-electrode EEG collector for noninvasive measurements of physiological signals, and EEG data were obtained for the prefrontal Fp1, Fpz, and Fp2 channels. Brain activity associated with emotions and potential changes in EEG features were linked to emotion regulation to objectively assess participants' emotional state.

3) System: The prototype experienced by Group A additionally recorded system data, including the *emotion score* calculated every 2 s in module A, the *relaxation index* generated every 2 s in module B, and the form and timing of triggering adaptive positive feedback in module C. These system data can be combined with physiological data for more detailed analysis and can further support insights into the feasibility of the positive feedback mechanism.

TABLE I
SUMMARY OF EEG RESTING-STATE RESULTS

Group	Baseline (Pre)			Intervention (Post)		
	Fp1	Fpz	Fp2	Fp1	Fpz	Fp2
A	1.37 ± 0.32	1.16 ± 0.30	1.34 ± 0.28	1.61 ± 0.29	1.25 ± 0.40	1.60 ± 0.29
B	1.60 ± 0.28	1.38 ± 0.26	1.55 ± 0.25	1.62 ± 0.32	1.38 ± 0.33	1.57 ± 0.31
p	0.027*	0.023*	0.025*	0.911	0.298	0.725

Note: Mean $\pm$ standard deviation; *significant at $p < 0.05$, **significant at $p < 0.01$.

D. Procedure

1) Stage 1: Baseline: Participants first completed self-report measures of VAS, STAI, and SWEMWBS. Subsequently, they wore three-electrode EEG VR device and closed their eyes to rest themselves for 1 min, to obtain the resting-state EEG data before the intervention as the baseline value.

2) Stage 2: Intervention: Once participants were ready to continue, they began to experience a 5-min VR mindfulness meditation intervention for the corresponding scenario: Group A experienced the prototype system with feedback, and Group B experienced the VR mindfulness meditation exercises without any feedback (mentioned in Section IV-A). Participants selected a VR scene between *flower companion* and *Yuhua river* that better suited their emotional relaxation based on their preference for sunny or rainy weather. The EEG signals of the participants were synchronously captured as task-state EEG data during the intervention (see Fig. 4). At the conclusion of the experience, participants in both groups were again subjected to a one-minute closed-eye resting-state measurement as the postintervention EEG data, and then removed the VR device and completed the postintervention self-report measures (VAS, STAI, and SWEMWBS).

3) Stage 3: Worry Induction: In order to measure the effectiveness of the intervention, participants undergo a worry induction task [49], [66], which is an effective 5–6 min. technique for triggering worry. Participants were instructed to consciously focus on nondepressive worry topics, which may be related to potential negative future events. After being alone for 5 min, participants were asked to rate their worry on a scale ranging from 0 (not at all) to 100 (extremely), including how catastrophic it would be, how likely it was to occur, and how well they would cope with it. Participants then completed the VAS, STAI, and SWEMWBS again.

4) Stage 4: Debrief: Participants were given the choice of whether or not to engage in a 2-min recentering mindfulness meditation to reduce stress, which was presented as an audio-only meditation. Semistructured interviews were then conducted with each participant to find out whether their personal emotion changes were within a reasonable range and what they felt more specifically during the VR mindfulness meditation experience.

VI. RESULTS

The results consisted of three main components: objective physiological data, subjective psychological data, and interviews. The EEG was used as a biomarker to support validity and engagement and was validated by fitting a linear regression model. Self-reported VAS, STAI, and SWEMWBS scales were used as psychometric measures, and quantitative analyses were completed by statistical methods. Paired samples t-tests were used to ascertain differences within groups, while independent samples t-tests were utilized to identify differences between groups. A significance level of 0.05 was chosen for all tests. Tables I and II present a summary of the findings from the physiological and self-report measures, respectively. Effectiveness is described based on the EEG, VAS, STAI, SWEMWBS, and recentering meditation selection rate. The EEG was used in conjunction with systematic data to verify the feasibility and acceptability of each module. The interviews were also summarized and combed for thematic clustering of qualitative feedback from participants.

TABLE II
SUMMARY OF SELF-REPORTED RESULTS

Variable	Group	Baseline	Intervention	Worry Induction
VAS	A	107.50 ± 74.39	49.17 ± 59.47	147.78 ± 80.08
	B	124.61 ± 71.81	50.72 ± 40.96	133.89 ± 58.83
	p	0.487	0.928	0.557
STAI	A	13.94 ± 3.92	9.72 ± 2.72	15.56 ± 4.00
	B	14.33 ± 4.30	10.44 ± 2.73	16.22 ± 4.04
	p	0.778	0.432	0.622
SWEMWBS	A	23.44 ± 2.48	25.44 ± 3.31	23.00 ± 3.53
	B	21.33 ± 3.99	23.56 ± 3.79	19.61 ± 5.14
	p	0.065	0.121	0.027*
Worry rating	A	–	–	53.67 ± 14.82
	B	–	–	57.83 ± 17.93
	p	–	–	0.453
Re-centering	A	–	–	0.22 ± 0.43
	B	–	–	0.56 ± 0.51
	p	–	–	0.041*

Note: Mean $\pm$ standard deviation; *significant at $p < 0.05$, **significant at $p < 0.01$.

A. Effectiveness

The EEG data were preprocessed through band-pass filtering (applying a 1–40 Hz band-pass filter to remove low-frequency drift and high-frequency noise), artifact removal (using independent component analysis to eliminate ocular artifacts), and rereferencing (employing average reference method). When analyzing the resting-state EEG data before and after the intervention, α/β was selected as an indicator of the relaxation state [70], and was calculated for each of the three channels Fp1, Fpz, and Fp2, with larger ratios indicating greater relaxation and a better emotional state. According to Table I, the baseline

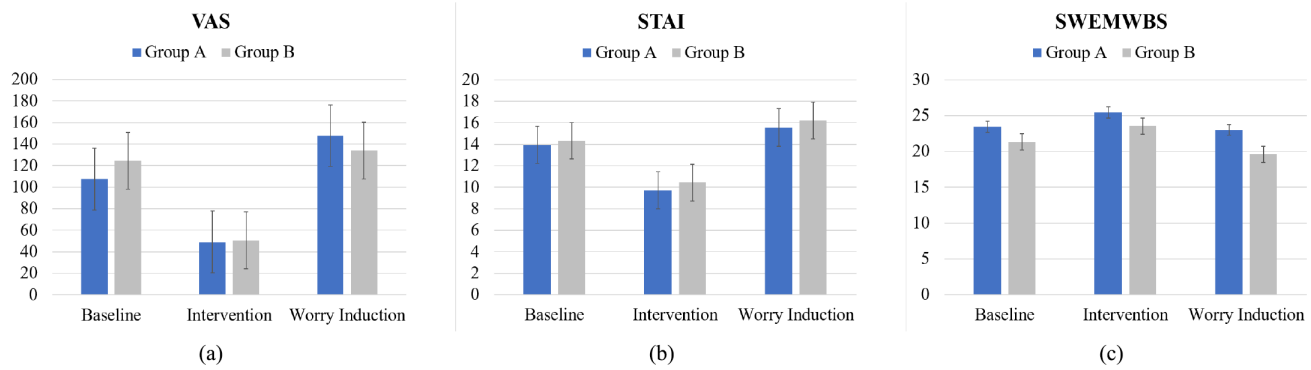

Fig. 5. Histogram of scale scores of experimental Group A and control Group B. (a) VAS. (b) STAI. (c) SWEMWBS.

value of Group A was significantly lower than Group B in the three channels (Fp1: $p = 0.027$*, Fpz: $p = 0.023$*, Fp2: $p = 0.025$*), whereas the intervention value had no significant difference between the two groups (Fp1, Fpz, and Fp2: $p > 0.05$), which indicated that Group A had poor baseline status but no difference from Group B after intervention. From preintervention to postintervention, there was a significant enhancement in both channels in Group A (Fp1: $p < 0.001$***, Fpz: $p = 0.124$, Fp2: $p < 0.001$***), whereas none of the channels in Group B showed significant changes (Fp1, Fpz, Fp2: $p > 0.05$). It can be seen that Group A was more relaxed. The amount of pre- and postintervention changes in Group A (Fp1: 0.24 ± 0.19, Fp2: 0.26 ± 0.23) was significantly greater than that in Group B (Fp1: 0.02 ± 0.38, Fp2: 0.02 ± 0.36) in both the Fp1 ($p = 0.038$*) and Fp2 ($p = 0.020$*) channels, verifying the importance of interventions containing intervention with positive feedback was significant in emotion regulation.

VAS [Fig. 5(a)] measures reported significant changes within both groups (Group A: baseline to intervention $p < 0.001$***, intervention to worry induction $p < 0.001$***; Group B: baseline to intervention $p < 0.001$***, intervention to worry induction $p < 0.001$***) but were not significant between the groups. STAI [Fig. 5(b)] self-reported anxiety scores were significantly lower from baseline to intervention in both groups (Group A: $p = 0.001$**; Group B: $p = 0.001$**), and then significantly increased to worry induction (Group A: $p < 0.001$***; Group B: $p < 0.001$***), with Group A having smaller values than Group B at all stages but this difference was not significant. Overall, both intervention modalities were effective in relieving anxiety, but the induced topic of worry still triggered anxiety, and comparatively Group A basically maintained a low level of anxiety at all stages.

Both groups produced significant changes in the indices measured by the SWEMWBS [Fig. 5(c)] between phases (Group A: baseline to intervention $p = 0.002$**, intervention to worry induction $p = 0.001$**; Group B: baseline to intervention $p = 0.001$**, intervention to worry induction < 0.001***), but overall the decline in mental well-being was more pronounced in Group B ($p = 0.017$*). Group A's mental well-being was better than Group B's at all stages and the difference after worry induction was significant ($p = 0.027$*), suggesting that the intervention with feedback gave more positive emotions and mental support. There was no significant difference in subjectively rated levels of worry between the two groups ($p = 0.453$), so participants' perceived ability to indicate worry was generally consistent. Group B, however, more clearly chose to engage in additional re-centering meditation ($p = 0.041$*), reflecting the fact that the intervention in Group A had a protective effect on mental health.

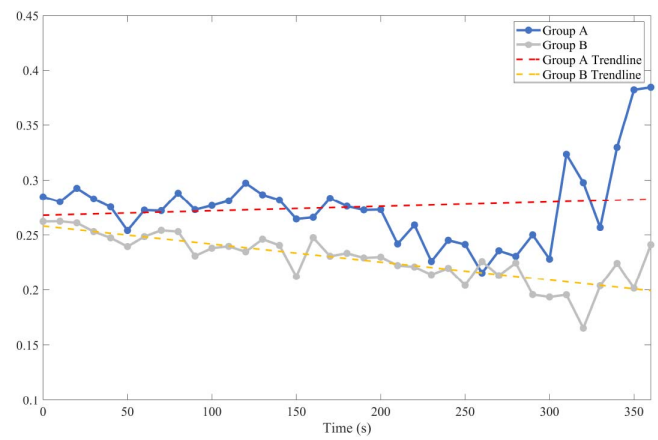

Fig. 6. Line chart and trend line of task state EEG in experimental Group A and control Group B.

For the EEG task-state data, we followed the *emotion score* formula mentioned in Section III-A, and used the results obtained from the combination of the three channels as an analytical index, and the normalized ratio (0–1) was calculated every 10 s to form a folded plot of emotion change based on the EEG task-state. Additionally, we conducted outlier detection and adjustment for EEG data. Given the relatively small sample size of the experiment, directly removing outliers could lead to significant loss of sample information. Therefore, we adjusted outliers by setting them to the upper or lower threshold values based on their magnitude. Meanwhile, the fitted linear regression model was used to assess the trend of participants' emotional change. As shown in Fig. 6, the fold line of Group A was located roughly above that of Group B, indicating a better emotional state during the intervention. The positive regression slope of Group A ($k = 4.071$e-05) indicated an upward trend in emotion scores, whereas the regression slope of Group B

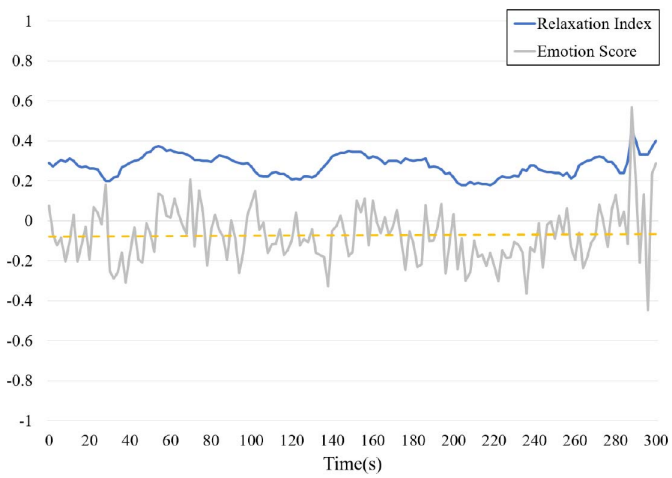

Fig. 7. Line chart of system data in experimental Group A.

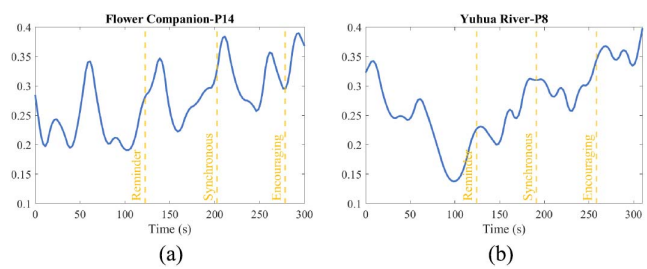

Fig. 8. Emotion change curves incorporating positive feedback for two participants in the experimental Group A. (a) Flower Companion-P14. (b) Yuhua River-P8.

was close to 0 ($k = -0.0001636$) indicating that the emotion tended to be stable as a whole, which stated that Group A was becoming more and more relaxed as the intervention proceeded, and the degree of emotion soothing to a greater extent.

B. Feasibility and Acceptability

The additional recorded *emotion score* and *relaxation index* of Group A were plotted as line graphs over time (Fig. 7), and a fitted linear regression was applied to the emotion scores to predict their trends. The *emotion score* curves were very volatile and often showed sudden increases or decreases, whereas the *relaxation index* curves were relatively stable and basically matched the linear regression line, which could be a better measure of the trend of emotion change and verified the feasibility of assessing relaxation based on the process.

In addition, the form and timing of the feedback recorded by the system were combined with the task-state EEG data of the participants in Group A to further observe the effects of different feedback on emotion. The type and time of feedback triggered by each participant varied, which was not suitable for statistical analysis of the whole group, so we selected one participant who triggered all the feedback from each of the two scenes of Group A (*Flower Companion*: P14, *Yuhua River*: P8) as a representative, so as to conduct specific analysis of individual cases. As shown in Fig. 8, the reminder feedback appeared at the lowest point of the emotion curves of both participants and then the curves immediately rose, reflecting the accurate timing of the reminder feedback triggering and its significant effect. As the dynamic effects associated with the synchronous feedback were presented, the participants became more and more relaxed. Near the end of the meditation process, the appearance of the encouragement feedback brought the emotion curve to a new peak. Overall, the participants' emotion continued to improve with the sequential triggering of the feedback, and the emotion states at the end of the process were better than those at the beginning, indicating that they had good acceptance of the adaptive feedback, and were able to achieve the goal of emotion regulation.

C. Interview Analysis

1) Reminder Feedback: Participants in Group A who triggered the reminder feedback ($n = 12$) all reported that it was helpful for them to focus their attention and could directly and clearly tell them how to adjust their state as soon as possible. "The voice reminder appeared in time when I was lost in thought and helped me to pull my attention back (P12)." "I thought it was a very thoughtful prompt, better than the purely audio meditations I have done before (P17)."

2) Synchronous Feedback: Some participants ($n = 10$) volunteered to mention the change brought by the synchronous feedback. In *Flower Companion*, participants noticed the small flowers around and "had an expectation of their blooming (P02)". They stated "cascading representation of the current state of mind (P13)" allowed for a better mood. In *Yuhua River*, participants were intrigued by the colored light spots, which they found to be a wonderful, positive guide. Some felt that the light spots created a "mutually reinforcing (P16)" connection between themselves and the scene. "When light spots converged into pillars of light, they suggested that my mind was moving from being scattered to being focused (P09)."

3) Encouraging Feedback: Those participants in Group A who were provided with encouraging feedback ($n = 6$) reported that the sound of insects and birds gave them inner peace (P14, P15). Participants who experienced the *Flower Companion* observed "a more relaxed feeling with the overall lighting up (of the scene) (P06)." Participants experiencing the *Yuhua River* saw the sunlight pouring down, with some finding it "a sense of surprise that the rain was over (P16)."

4) EEG-Based Positive Feedback Mechanism: After we explained the feedback mechanism to Group A, they reported that the feedback provided was able to "make people think more positively, and the changes in the scene made people more attentive (P06)" and that it could be "adjusted in real time according to the emotion (assessed by EEG), and was not static (P11)," and is "tailored to the individual's feelings and is highly targeted (P03)." On the other hand, the participants in Group B generally reported that they were observing the scene as an object during the experience ($n = 7$). Although they did not get bored, they still looked forward to richer dynamic changes and interactions with the scene. Some participants expressed

the need for timely feedback in order to understand "how far the meditation has gone, what condition they are in, and how to adjust next (P26)."

VII. Discussion

The main purpose of this experiment was to investigate the effect of EEG-based positive feedback mechanisms applied to VR mindfulness meditation on emotion regulation. The results showed favorable evidence of objective physiological and subjective psychological compared to our control group without any feedback. After experiencing the enhanced intervention containing feedback, physiological data from the experimental sample showed an increase in relaxation by almost 20%, a decrease in self-reported mean anxiety by 54.3%, and little or no effect of the induced topic of worry on good psychological performance. Positive comments and offers from the subjects reflect a willingness to integrate VR mindfulness meditation with EEG-based positive feedback.

A. Effectiveness

The experimental results showed that the intervention combined with EEG-based positive feedback had a significant advantage in emotion regulation. Compared to the control group, the amount of change in the resting state on both Fp1 and Fp2 channels before and after the intervention reflected an increase in relaxation by more than 90%, and the emotional line graph obtained through the task state reflected a consistently good and improving emotional state during the intervention, as well as self-reported lower levels of anxiety and higher levels of mental health after the intervention. After the worry induction phase, the experimental group showed a 74.4% increase in the protection of the psyche against the topic of worry, and there was a 34% reduction difference in the selection rate of additional self-resetting meditations. According to the conceptual model of Human-Engaged Computing (HEC) [71], [72], it is known that the user and the system form a closed-loop of positive interaction through the EEG positive feedback mechanism, and the user's physiological data participate in the system's computation and generate positive dynamic changes, which are then fed back to the user to further soothe their emotions, thus achieving the purpose of human-computer synergistic interaction and mutual promotion. It not only optimizes the emotion regulation effect of VR mindfulness meditation, but also improves the user's own emotion regulation ability.

B. Feasibility

Our proposed process-based relaxation assessment module is feasible. As can be seen through the system data line graph, the *emotion score* curve cannot truly reflect the user's emotion changes, and if real-time feedback is based on the emotion state at a certain moment, the dynamic changes presented are very volatile and out of control, which is not in line with the expected effect. The *relaxation index* curve can basically accurately assess the trend of emotion change, while retaining more details of the change compared with the predicted trend line, which plays a certain role in smoothing the dynamic effect of feedback. In conclusion, the comprehensive assessment of the user's relaxation level based on the emotional state at a certain time can reflect the trend of emotional change more comprehensively and realistically. Using the *relaxation index* as the feedback indicator can ensure that the feedback provided is more precise and stable.

C. Acceptability

The acceptability of the adaptive positive decision feedback module have been demonstrated. Mindfulness is a balance between relaxation and concentration [73], and our proposed feedback approach can effectively help users maintain this balance and stay in a positive state, resulting in a mind-flow experience. Two of our selected participants' emotional scores about mindfulness gradually decreased when they were in the initial stage of emotion regulation, which was attributed to the fact that free breathing made them overly relaxed and mind-wandering. At this time, providing breathing guidance could help them to be more focused. However, focusing for a period of time may cause stress, and at this time, providing new stimuli inducing positive emotions could allow them to relax again. Meanwhile, participants in the experimental group expressed positive perceptions of the three forms of feedback. Reminder feedback was the most intuitive, with timely voice prompts to help focus attention and clear guidance to make the task easier. Synchronous feedback presents the user's mental state with subtle changes, increases attention to self-perception, allows users to regulate their state more consciously, and improves user engagement. Encouraging feedback provides positive emotional induction to further promote emotional relief, and adding new variations to the scene can effectively reduce the shedding rate. Diversified forms of feedback provide a richer interactive experience that meets the needs of more people.

D. Limitations and Future Work

First, the source of the physiological data we used is singular. In our experimental environment, EEG signal is often interfered with, making it even more challenging to integrate it into real-world application environments. In future work, other physiological indicators reflecting emotions, such as respiration and heart rate, can be incorporated. The fusion of multimodal physiological information can depict emotions more comprehensively, improving the accuracy and robustness of emotion recognition. Second, current feedback mechanisms primarily focus on alterations in audiovisual elements within virtual environments, whereas olfactory and tactile stimuli from real-world settings are employed as constants to enhance immersion. Next, we aim to incorporate olfactory and tactile interfaces into the feedback mechanism, combining a richer set of feedback variables with biometric data to enhance effectiveness. Finally, substantial work remains to ascertain the effect of EEG-based positive feedback mechanism within diagnostic populations.

In the future, we will further explore how these findings can be translated into clinical applications and conduct long-term longitudinal experiments across diverse populations.

VIII. Conclusion

In this article, we propose an EEG-based positive feedback mechanism applied to VR mindfulness meditation aimed at enhancing emotion regulation. This mechanism can promptly respond to the users' EEG data, synchronously adjust the output of audio–visual stimuli, and establish an adaptive positive closed-loop system. Specifically, the process-based relaxation assessment provides a comprehensive measure of emotional changes, improving the accuracy and reliability of emotional quantification. The adaptive positive decision feedback is tailored to the goals and needs of emotion regulation, ensuring precise and satisfactory implementation of feedback strategies. Experimental results indicate that incorporating this mechanism into VR mindfulness meditation can effectively alleviate anxiety and promote relaxation within a short period, reducing the intrusion of negative thoughts. We found that personalized dynamic effects enhance immersion and improve user experience, while diversified feedback forms offer appropriate guidance to help users maintain mindfulness. Overall, the EEG-based positive feedback mechanism significantly increases user engagement, optimizes the effects of emotion regulation, and plays a certain protective role in mental health. This study offers valuable insights into the integration of biofeedback with VR mindfulness meditation and presents significant opportunities to improve both the efficacy and engagement levels associated with emotion regulation strategies.

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